Jose Antonio Elizalde

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Education

University of California, Los Angeles **Sept 2025 – June 2027**
B.S. in Computer Science and Engineering - GPA: 3.5 *Westwood, CA*

Experience/Research

REUSE Research Program – Carnegie Mellon University **May 2025 – Aug 2025**
Undergraduate Researcher, Human-Computer Interaction Institute (HCII) *Pittsburgh, PA*

- Built an interactive application studying whether **gamified in-ride interactions** can improve communication and empathy between rideshare drivers and passengers, refining its features through usability testing across **nine workshops**.
- Co-authored *Beyond Riding: Passenger Engagement with Driver Labor through Gamified Interactions*, accepted to **CHI 2026** (ACM Conference on Human Factors in Computing Systems).

SURE C² Research Program – University of California, Los Angeles **Aug 2024 – May 2025**
Undergraduate Researcher, Department of Mathematics *Los Angeles, CA*

- Researched transfer systems on partially ordered sets, discovering a recursive pattern on X_n^{++} and co-authoring a paper presented at UT Arlington; wrote a **C++** program to automate counting all transfer systems on X_n^{++}.

Projects

DailyCodeForce | ***Next.js, TypeScript, PostgreSQL, Prisma, Go, Docker, Nginx*** **Aug 2026 – Present**

- Built a self-hosted platform serving **4 Codeforces problems daily** (one per difficulty tier) with C++ reference solutions, run as **7 Docker Compose services** (Next.js, PostgreSQL, sandboxed judge, Go Discord bot, presence tracker, cron scheduler, **Nginx**) on **DigitalOcean**.
- Built a **sandboxed C++17 judge** executing untrusted submissions in **under 1s** under **four enforced rlimits** (256MB address space, 3s CPU time, 64 processes, 10MB writes) inside a hardened container (pids cap, `noexec` /tmp, no-new-privileges).
- Implemented scheduling with a **90-day deduplication window** and **30-day automated backfill** over a **113-problem** PostgreSQL pool (**Prisma**), delivering four new problems daily with **zero repeats**.

Robinhood Discord Portfolio Bot | ***C++, Python, FastAPI, pandas, NumPy*** **June 2026 – Present**

- Built a **C++** Discord bot (D++) and Python **FastAPI** backend that report live Robinhood portfolio analytics, recommend stocks, and render a 3D graph of portfolio risk, serving requests at **486ms median latency** and **783ms p99** across 200 sampled requests on a **single-vCPU 512MB** Linux VPS.
- Architected a **decoupled two-service system** with pickled session persistence and proactive token refresh, restoring sessions across restarts without re-login; fully available **2s after cold start**.
- Built a two-stage screening pipeline evaluating all **503 S&P 500 tickers** on momentum in **39s** before applying fundamental filters, running as a **background thread** within a **107MB peak memory footprint** under **systemd**.

Jump The Gun – Game Director | UCLA ACM Studio | ***Unity, C#, Git*** **Oct 2025 – Present**

- Led the design and development of *Jump the Gun*, a single-player first-person action shooter, defining gameplay mechanics and coordinating implementation across an **8-person team** of programmers, artists, and designers.
- Implemented a custom **physics-based movement controller** in **C#**, replacing Unity's default character controller with manually programmed movement and collision logic.
- Ran the team's **Git branching and pull-request review** workflow for features built by multiple contributors.

U.F. ZOO | ***Unity, C#, Blender*** **– Gameplay Programmer, team of 20** **Aug 2026 – Present**

- Build **gameplay systems** for a city-builder/deck-builder roguelike tycoon in which players build a zoo in the sky and clear point quotas, targeting a **Steam release in January 2027**.
- Implemented the **card ability system**, covering **15+ distinct card effects** and their resolution rules, plus the **multi-island expansion** feature that lets players grow their zoo outward.

Extracurriculars

UCLA ACM Studio – Intern **Sep 2025 – Present**

- Help run workshops that teach new and returning members game development topics such as building a roguelike from scratch and 3D modeling.
- Help organize the club's internal **game jams**, form teams for external jams, and answer members' technical and logistical questions throughout.

Technical Skills

Languages: C++, C, C#, Python, Go, TypeScript, SQL, Bash
Frameworks & Infra: Next.js, React, FastAPI, PostgreSQL, Prisma, Docker, Nginx, Linux, systemd, DigitalOcean
Tools: Git/GitHub, CMake, Unity, Blender, Arduino, pandas, Vim/NeoVim, LaTeX